Supplementary material

Maternal mortality in Córdoba, Colombia, 2015 to 2024

The figures and tables in this supplement are computed from the derived data included in the reproducibility package; the place-of-occurrence table (Table S9) and the characteristics table (Table S12) draw on individual-level records that are not redistributed.

Table S1. Counts of maternal death under the two non-equivalent case definitions

Source and definition	ICD-10 codes or criterion	Role in the analysis	n
Vital statistics, maternal death	O00 to O95, O98 to O99 and A34, plus O96 and O97	Universe of maternal deaths in the civil registry	242
Vital statistics, early maternal death	O00 to O95, O98 to O99 and A34	Numerator of the mortality ratio	162
Vital statistics, late maternal death	O96	Excluded from the ratio; analyzed as a series of its own and in the sensitivity analysis	78
Surveillance, notified maternal death	Notification criterion, not restricted to chapter XV	Reported for comparison of totals	247

Sources: national vital statistics and national surveillance microdata. The two systems apply different case definitions: the surveillance protocol covers a broader set of notifiable maternal deaths. The totals are shown side by side for reference and the proximity between 242 and 247 is not evidence of agreement at the case level, which cannot be assessed because the records carry no common identifier. The flow of records from each source to the analytic counts is shown in Supplementary Figure S2. Two deaths coded O97 (death from sequelae of obstetric causes one year or more after termination) are part of the 242-death registry universe but are not maternal deaths within the first year and are excluded from both the early and the late series.

Table S2. Annual maternal mortality ratio, department and rest of the country

Year	Deaths	Live births	Ratio, department (95% CI)	Deaths, rest	Live births, rest	Ratio, rest (95% CI)	Rate ratio, log-normal (95% CI)	Rate ratio, exact conditional (95% CI)
2015	23	26 295	87.5 (55.4 to 131.2)	329	634 398	51.9 (46.4 to 57.8)	1.69 (1.11 to 2.57)	1.69 (1.05 to 2.57)
2016	17	25 919	65.6 (38.2 to 105.0)	314	621 154	50.6 (45.1 to 56.5)	1.30 (0.80 to 2.11)	1.30 (0.75 to 2.11)
2017	18	26 668	67.5 (40.0 to 106.7)	303	628 770	48.2 (42.9 to 53.9)	1.40 (0.87 to 2.25)	1.40 (0.82 to 2.25)
2018	18	26 239	68.6 (40.7 to 108.4)	266	618 785	43.0 (38.0 to 48.5)	1.60 (0.99 to 2.57)	1.60 (0.93 to 2.57)
2019	12	25 004	48.0 (24.8 to 83.8)	294	606 064	48.5 (43.1 to 54.4)	0.99 (0.56 to 1.76)	0.99 (0.51 to 1.75)
2020	19	24 767	76.7 (46.2 to 119.8)	389	594 835	65.4 (59.1 to 72.2)	1.17 (0.74 to 1.86)	1.17 (0.70 to 1.86)
2021	26	25 975	100.1 (65.4 to 146.7)	476	582 108	81.8 (74.6 to 89.5)	1.22 (0.82 to 1.82)	1.22 (0.79 to 1.82)
2022	15	24 207	62.0 (34.7 to 102.2)	255	543 020	47.0 (41.4 to 53.1)	1.32 (0.78 to 2.22)	1.32 (0.73 to 2.22)
2023	7	21 080	33.2 (13.4 to 68.4)	225	487 219	46.2 (40.3 to 52.6)	0.72 (0.34 to 1.53)	0.72 (0.29 to 1.51)
2024	7	17 625	39.7 (16.0 to 81.8)	190	425 523	44.7 (38.5 to 51.5)	0.89 (0.42 to 1.89)	0.89 (0.35 to 1.87)
2015-2024	162	243 779	66.5 (56.6 to 77.5)	3 041	5 741 876	53.0 (51.1 to 54.9)	1.25 (1.07 to 1.47)	1.25 (1.06 to 1.47)

Exact Poisson intervals for the ratios. The rate ratio is given with the usual log-normal interval and with an exact conditional interval, which does not rely on the normal approximation and is preferable on annual counts of 7 to 26 deaths; the two agree on which years separate from the comparator. The comparator comprises the other 32 territorial units of the 33 into which the country is divided, excluding the department, so the two groups are independent. Rates are per 100 000 live births.

Table S3. Stability of the trend estimates when one year is omitted

Series	Year omitted	Annual percent change	p
All early deaths	Full series	-7.0%	0.023
	Omitting 2015	-5.8%	0.114

Series	Year omitted	Annual percent change	p
	Omitting 2016	-7.6%	0.020
	Omitting 2017	-7.1%	0.025
	Omitting 2018	-7.0%	0.025
	Omitting 2019	-6.8%	0.026
	Omitting 2020	-7.4%	0.018
	Omitting 2021	-7.4%	0.018
	Omitting 2022	-8.8%	0.013
	Omitting 2023	-5.6%	0.103
	Omitting 2024	-6.9%	0.055
Direct causes	Full series	-7.1%	0.061
	Omitting 2015	-6.3%	0.160
	Omitting 2016	-6.1%	0.136
	Omitting 2017	-7.1%	0.063
	Omitting 2018	-7.1%	0.057
	Omitting 2019	-6.6%	0.072
	Omitting 2020	-7.1%	0.061
	Omitting 2021	-7.1%	0.061
	Omitting 2022	-8.0%	0.059
	Omitting 2023	-6.9%	0.111
	Omitting 2024	-8.5%	0.058
Indirect causes	Full series	-6.2%	0.251
	Omitting 2015	-3.4%	0.598
	Omitting 2016	-10.2%	0.060
	Omitting 2017	-6.1%	0.275
	Omitting 2018	-5.9%	0.301
	Omitting 2019	-6.6%	0.241
	Omitting 2020	-7.4%	0.185
	Omitting 2021	-7.4%	0.185
	Omitting 2022	-9.9%	0.124
	Omitting 2023	-2.6%	0.661
	Omitting 2024	-3.1%	0.618
Late maternal deaths	Full series	+10.6%	0.013
	Omitting 2015	+8.8%	0.065
	Omitting 2016	+8.8%	0.050

Series	Year omitted	Annual percent change	p
	Omitting 2017	+12.8%	0.007
	Omitting 2018	+10.4%	0.016
	Omitting 2019	+10.8%	0.012
	Omitting 2020	+11.3%	0.008
	Omitting 2021	+11.3%	0.008
	Omitting 2022	+10.2%	0.023
	Omitting 2023	+11.3%	0.016
	Omitting 2024	+10.8%	0.033

Poisson model with live births as offset and a pandemic-period indicator, refitted omitting one calendar year at a time. Omitting 2020 and omitting 2021 give the same slope because a single remaining pandemic year is absorbed entirely by the indicator and contributes nothing to the calendar-year term. The point estimate of each series stays within a narrow band; the p value of the direct-cause series stays above 0.05 in all ten refits (0.057 to 0.160), which is why its decline is reported as a modeled long-term trend and not as an established year-on-year reduction. The indirect-cause series has no trend in any refit.

Table S4. Early and late maternal deaths, and the joint model of the two death windows

Year	Early deaths	Late deaths	Live births	Early ratio (95% CI)	Late ratio (95% CI)
2015	23	4	26 295	87.5 (55.4 to 131.2)	15.2 (4.1 to 38.9)
2016	17	4	25 919	65.6 (38.2 to 105.0)	15.4 (4.2 to 39.5)
2017	18	10	26 668	67.5 (40.0 to 106.7)	37.5 (18.0 to 69.0)
2018	18	7	26 239	68.6 (40.7 to 108.4)	26.7 (10.7 to 55.0)
2019	12	9	25 004	48.0 (24.8 to 83.8)	36.0 (16.5 to 68.3)
2020	19	11	24 767	76.7 (46.2 to 119.8)	44.4 (22.2 to 79.5)
2021	26	4	25 975	100.1 (65.4 to 146.7)	15.4 (4.2 to 39.4)
2022	15	11	24 207	62.0 (34.7 to 102.2)	45.4 (22.7 to 81.3)
2023	7	9	21 080	33.2 (13.4 to 68.4)	42.7 (19.5 to 81.0)
2024	7	9	17 625	39.7 (16.0 to 81.8)	51.1 (23.3 to 96.9)
2015-2024	162	78	243 779	66.5 (56.6 to 77.5)	32.0 (25.3 to 39.9)

Terms of the joint model of the two death windows

Term of the joint model	Estimate (95% CI)	p
Calendar year, early deaths	-7.0% per year (-12.7 to -1.0)	0.023
Death window by calendar year	+19.0% per year (+7.5 to +31.6)	<0.001
Pandemic period, early deaths	rate ratio 1.68 (1.16 to 2.44)	0.006
Death window by pandemic period	rate ratio 0.48 (0.24 to 0.94)	0.033
Implied slope, early deaths	-7.0% per year	
Implied slope, late deaths	+10.6% per year	
Implied pandemic-period rate ratio, late deaths	0.81	

Rates are per 100 000 live births with exact Poisson intervals. The joint model is fitted to the 20 annual counts of the two windows, with live births as offset, and includes the death window, the calendar year, the pandemic-period indicator and the interaction of the window with each of the last two. The two interaction terms answer different questions: the first, whether the two windows follow different secular slopes; the second, whether the pandemic period acted on them differently. Both are needed, because a model with only the first would force the two windows to share the same pandemic effect.

Table S5. Dispersion and quasi-Poisson refits of the trend models

Series	Dispersion	Annual change, Poisson (95% CI)	Annual change, quasi-Poisson (95% CI)	Pandemic period, Poisson (95% CI)	Pandemic period, quasi-Poisson (95% CI)
All early deaths	0.57	-7.0% (-12.7 to -1.0)	-7.0% (-12.7 to -1.0)	1.68 (1.16 to 2.44)	1.68 (1.16 to 2.44)
Direct causes	0.38	-7.1% (-14.0 to +0.3)	-7.1% (-14.0 to +0.3)	0.78 (0.43 to 1.41)	0.78 (0.43 to 1.41)
Indirect causes	1.66	-6.2% (-16.0 to +4.7)	-6.2% (-21.0 to +11.2)	3.60 (2.11 to 6.13)	3.60 (1.57 to 8.23)
Late maternal deaths	1.04	+10.6% (+2.2 to +19.7)	+10.6% (+0.3 to +21.9)	0.81 (0.46 to 1.42)	0.81 (0.40 to 1.62)

Joint models with interaction terms

Interaction term	Dispersion	Poisson, rate ratio (95% CI)	Quasi-Poisson, rate ratio (95% CI)
Cause group by year	1.02	1.01 (0.88 to 1.15)	1.01 (0.87 to 1.17)
Cause group by pandemic period	1.02	4.64 (2.09 to 10.29)	4.64 (1.92 to 11.17)
Death window by year	0.81	1.19 (1.08 to 1.32)	1.19 (1.08 to 1.32)
Death window by pandemic period	0.81	0.48 (0.24 to 0.94)	0.48 (0.24 to 0.94)

Dispersion is the Pearson statistic divided by its degrees of freedom. The quasi-Poisson scale is applied only when the estimated dispersion exceeds one, using the t distribution; where the dispersion is at or below one the Poisson model is kept unchanged, because a dispersion below one on ten annual observations is compatible with chance and does not justify narrower intervals. The series with dispersion above one, the indirect causes (1.66), the late maternal deaths (1.04) and the joint cause-group model (1.02), are refitted and their intervals widen accordingly; the other series, with dispersion at or below one, are unchanged. The point estimates are identical by construction. The second part of the table repeats the four interaction terms of the joint models under both specifications. No conclusion of the paper changes under either.

Table S6. Global Moran’s I under different inputs (999 permutations)

Input	Moran's I	Expected I	p	Use
Crude municipal rate	-0.007	-0.034	0.37	Primary
Empirical Bayes adjustment for population size	+0.039	-0.034	0.26	Sensitivity: corrects the bias of the index on rates with unequal denominators
Global empirical Bayes smoothing	+0.069	-0.034	0.20	Sensitivity
95th percentile of the permutation distribution	+0.168			Detection threshold

Queen contiguity weights, row-standardized, on the 30 municipalities. The adjusted index follows the empirical Bayes correction for heterogeneous population sizes, which addresses the known bias of the index when it is computed on rates with very unequal denominators. Local indicators of spatial association identified no municipality with p below 0.05 on the crude rates.

Table S7. Ecological association with municipal poverty and statistical power

Analysis	Rate ratio per 10 index points (95% CI)	p
Poisson, all 30 municipalities	1.00 (0.90 to 1.11)	0.99
Poisson, excluding the departmental capital	1.00 (0.86 to 1.17)	0.98
Power for a true rate ratio of 1.05	16.3%	
Power for a true rate ratio of 1.10	47.2%	
Power for a true rate ratio of 1.15	79.6%	
Power for a true rate ratio of 1.20	95.7%	

Power estimated by simulation with 3 000 replicates preserving the 30 municipalities, the observed live births, the observed distribution of the poverty index and the 162 observed deaths. The design reaches 80% power at a rate ratio of about 1.15 per 10 index points; smaller associations would not be detected reliably with these data.

Table S8. Segmented temporal trend of the maternal mortality ratio

Period	Years	Deaths	Annual percent change (95% CI)	p
Pre-pandemic	2015-2019	88	-10.1% (-21.9 to +3.7)	0.143
Pandemic	2020-2021	45	+33.7% (+2.6 to +74.1)	0.031
Post-pandemic	2022-2024	29	-32.7% (-48.0 to -12.9)	0.003
Whole period, single slope	2015-2024	162	-7.0% (-12.7 to -1.0)	0.023

Poisson model with live births as offset and slope changes at 2019 and 2021, fixed a priori on the calendar of the pandemic and not estimated from the data. The knots remain at 2019 and 2021; the years in the table are the non-overlapping blocks used to count the deaths, which therefore add up to 162. Counts per block are shown because each slope rests on few events. The segmented model is exploratory: it estimates four parameters from ten annual observations. The three slopes are the lines fitted over 2015 to 2019, 2019 to 2021 and 2021 to 2024.

Table S9. Place of occurrence by subregion of residence

Subregion of residence	Municipalities	Deaths (denominator)	Occurred in the departmental capital, n	Occurred in the capital, %	Occurred outside the department, n
Medio Sinú	5	67	62	92.5	1
Alto Sinú	2	15	13	86.7	1
San Jorge	8	32	24	75.0	3
Costanera	6	16	12	75.0	0
Sabanas	4	15	5	33.3	6
Bajo Sinú	5	17	5	29.4	1

Subregion of residence	Municipalities	Deaths (denominator)	Occurred in the departmental capital, n	Occurred in the capital, %	Occurred outside the department, n
All subregions	30	162	121	74.7	12

Deaths are attributed by municipality of residence throughout the analysis; this table describes where they occurred, using the department and municipality of occurrence recorded in the same death certificate. Numerators and denominators are given as counts: of the 113 women who did not live in the departmental capital, 73 died there (64.6%); the corresponding figure including the 48 capital residents who also died there, the figure is 121 of 162. The subregion of residence follows the departmental administrative division and the correspondence is given in Table S10. This table describes where the events occurred; it carries no information on travel distance, on the distribution of services by level of care, on referral episodes or on clinical times, and it is not evidence about the timeliness or the quality of the care received.

Table S10. Correspondence between municipality of residence and subregion

Subregion	Municipalities	DANE codes
Alto Sinú	Tierralta, Valencia	23807, 23855
Bajo Sinú	Chima, Cotorra, Lorica, Momil, Purisima	23168, 23300, 23417, 23464, 23586
Costanera	Canalete, Los Cordobas, Moñitos, Puerto Escondido, San Antero, San Bernardo del Viento	23090, 23419, 23500, 23574, 23672, 23675
Medio Sinú	Cerete, Cienaga de Oro, Monteria, San Carlos, San Pelayo	23001, 23162, 23189, 23678, 23686
Sabanas	Chinu, Sahagun, San Andres de Sotavento, Tuchin	23182, 23660, 23670, 23815
San Jorge	Ayapel, Buenavista, La Apartada, Montelibano, Planeta Rica, Pueblo Nuevo, Puerto Libertador, San Jose de Ure	23068, 23079, 23350, 23466, 23555, 23570, 23580, 23682

Division of the department into its six subregions, as used in Tables S9 and S12. The codes are those of the national statistical office. The 30 municipalities cover the whole department.

Table S11. Indirect standardization of the departmental excess for maternal age

Maternal age group	Deaths, Córdoba	Rate, Córdoba	Rate, rest of the country	Expected in Córdoba
10-14	2	71.0	62.4	1.8
15-19	17	31.6	39.5	21.3
20-24	32	44.2	38.7	28.0
25-29	46	81.7	46.6	26.2
30-34	34	94.2	60.7	21.9
35-39	22	125.3	96.7	17.0
40-44	8	181.4	152.8	6.7
45-49	1	318.5	258.2	0.8
Total	162			123.7

Rates per 100 000 live births. Expected deaths apply the age-specific rate of the rest of the country to the age structure of live births in Córdoba. The standardized mortality ratio is 162 / 123.7 = 1.31 (95% CI 1.12 to 1.53), close to the crude rate ratio of 1.25, so the departmental excess is not explained by the age distribution of mothers, in eight five-year groups from 10 to 49. Maternal age was recorded in all 162 deaths.

Table S12. Characteristics of the 162 early maternal deaths

Characteristic	n (%)
Age group, years	
10 to 14	2 (1.2)
15 to 19	17 (10.5)
20 to 24	32 (19.8)

Characteristic	n (%)
25 to 29	46 (28.4)
30 to 34	34 (21.0)
35 to 39	22 (13.6)
40 to 44	8 (4.9)
45 to 49	1 (0.6)
Insurance regime (not stated: 1, 0.6%)	
Subsidized	126 (77.8)
Contributory	30 (18.5)
Uninsured	5 (3.1)
Area of residence	
Municipal seat	92 (56.8)
Populated center	24 (14.8)
Dispersed rural	46 (28.4)
Subregion of residence	
Medio Sinú	67 (41.4)
San Jorge	32 (19.8)
Bajo Sinú	17 (10.5)
Costanera	16 (9.9)
Sabanas	15 (9.3)
Alto Sinú	15 (9.3)
Ethnic self-identification	
None of the above	147 (90.7)
Afro-descendant	10 (6.2)
Indigenous	5 (3.1)
Educational level (not stated: 49, 30.2%)	
None	3 (1.9)
Preschool	3 (1.9)
Primary	24 (14.8)
Secondary	33 (20.4)
Upper secondary, academic	38 (23.5)
Upper secondary, technical	2 (1.2)
Technical professional	3 (1.9)
University	7 (4.3)
Marital status (not stated: 21, 13.0%)	
Consensual union, 2 years or more	73 (45.1)
Consensual union, less than 2 years	24 (14.8)
Single	24 (14.8)
Married	16 (9.9)

Characteristic	n (%)
Separated or divorced	3 (1.9)
Widowed	1 (0.6)
Pregnant at the time of death	
Yes	35 (21.6)
No	127 (78.4)
Place of death	
Health institution	153 (94.4)
Outside a health institution	9 (5.6)
Medical assistance received	
Yes	154 (95.1)
No	8 (4.9)
Death occurred in Córdoba	
Yes	150 (92.6)
No	12 (7.4)
Period	
2015 to 2021	133 (82.1)
2022 to 2024	29 (17.9)

Source: national vital statistics. Percentages are over the 162 deaths. Where a field was not stated in some records, the count is noted beside the variable; variables without such a note were complete. This table corresponds to the sociodemographic description in the Results section.

Table S13. Crosswalk of ICD-10 codes to cause groups

Cause group	ICD-10 codes observed	Deaths
Hypertensive disorders	O13, O141, O142, O149, O150, O152, O16	35
Obstetric hemorrhage	O441, O459, O469, O710, O711, O712, O720, O721, O723	20
Abortive outcomes	O009, O021, O061, O063, O065	14
Pregnancy-related infection	O85, O862	8
Obstetric embolism	O882	5
Other direct obstetric causes	O225, O25, O410, O411, O669, O754, O759, O903, O904, O909	11
Indirect causes	O240, O241, O981, O985, O986, O987, O988, O990, O991, O993, O994, O995, O996, O998	68
Undetermined causes	O95	1
Total		162

Assignment follows the direct and indirect distinction of the WHO application of ICD-10 (ICD-MM): pre-existing diabetes (O24.0 to O24.3) is counted as indirect, codes O98 and O99 as indirect, and the unspecified obstetric death (O95) as undetermined, neither direct nor indirect. All 162 early-death codes are listed and every observed code was assigned to a group.

Table S14. STROBE checklist for this report

Item	Recommendation	Location
1 Title and abstract	Design in the title; structured summary	Title; Abstract
2 Background	Scientific background and rationale	Introduction
3 Objectives	Specific objectives	Introduction, last paragraph

Item	Recommendation	Location
4 Study design	Key elements of the design	Methods, Design and data sources
5 Setting	Setting, locations, periods	Methods, Design and data sources
6 Participants	Eligibility and case selection	Methods, Definitions; Figure S2
7 Variables	Outcomes, exposures, predictors	Methods, Definitions and Statistical analysis
8 Data sources	Sources and measurement	Methods, Design and data sources
9 Bias	Efforts to address sources of bias	Methods; Limitations
10 Study size	How the study size was arrived at	Methods; Results (counts by year)
11 Quantitative variables	Handling of quantitative variables	Methods, Statistical analysis
12 Statistical methods	All methods, subgroups, sensitivity	Methods, Statistical analysis
13 Participants	Numbers at each stage	Figure S2; Table S1
14 Descriptive data	Characteristics of participants	Table S12
15 Outcome data	Numbers of outcome events	Results; Tables 1 and 2
16 Main results	Estimates and confidence intervals	Results; Table 2
17 Other analyses	Subgroups, sensitivity, interactions	Results; Tables S3 to S11
18 Key results	Summary with reference to objectives	Discussion, first paragraph
19 Limitations	Limitations and direction of bias	Limitations
20 Interpretation	Cautious overall interpretation	Discussion; Conclusions
21 Generalizability	External validity	Discussion; Limitations
22 Funding	Role of funders	Title page

Checklist for cross-sectional/ecological reporting under STROBE. Section names refer to the main manuscript and to this supplement.

Supplementary figures

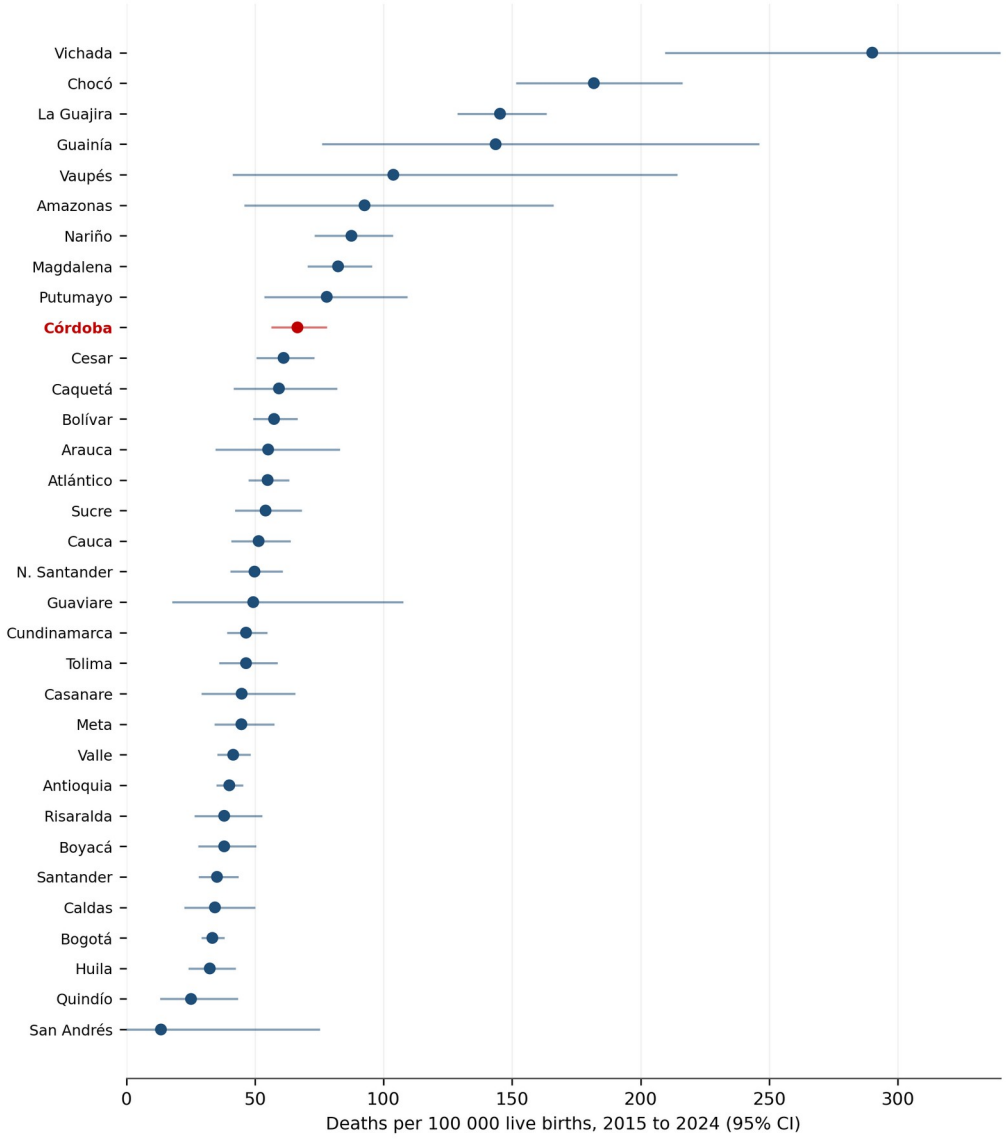
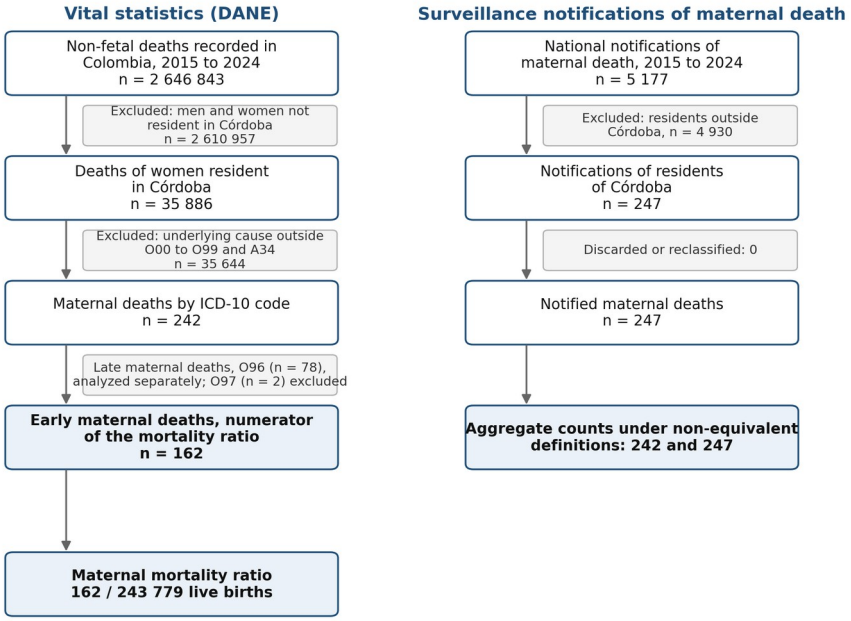
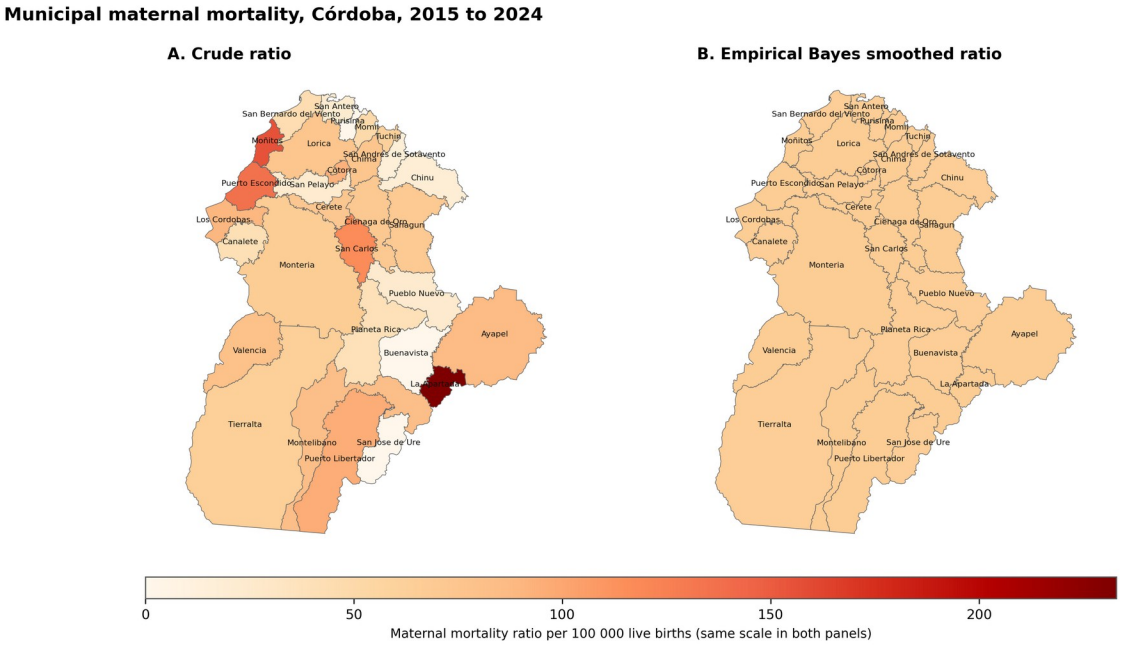


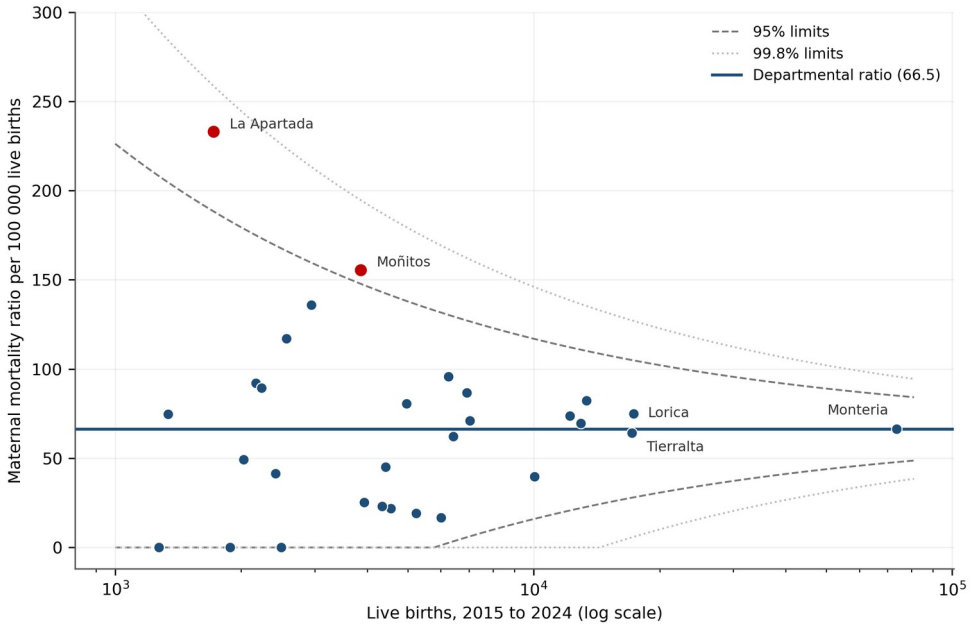
Figure S1. Ranking of the 33 territories by the pooled maternal mortality ratio, 2015 to 2024. The department under study is highlighted. Bars are exact Poisson confidence intervals. Source: national vital statistics.



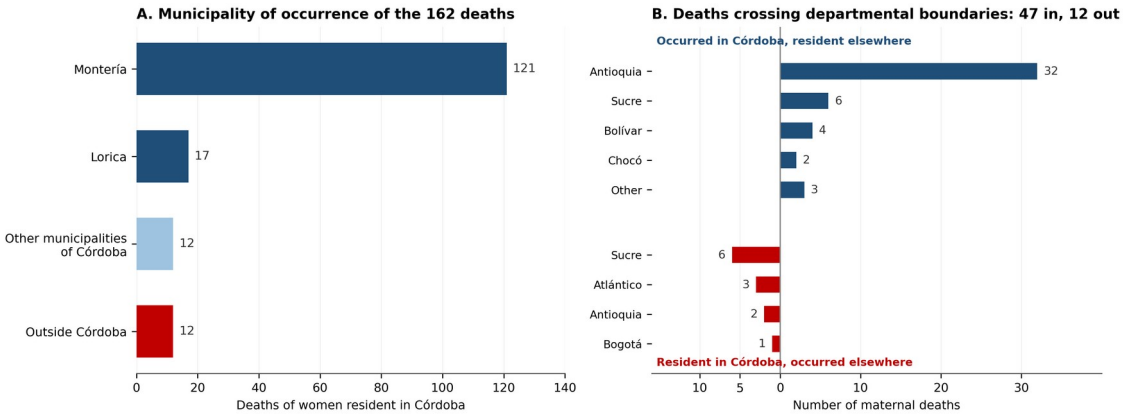
Supplementary Figure S2. Flow of records from the two national sources to the analytic counts, Córdoba, Colombia, 2015 to 2024. Vital statistics provide the numerator of the mortality ratio; surveillance notifications are counted under a different case definition and are reported for comparison of totals only.



Supplementary Figure S3. Municipal maternal mortality ratio, Córdoba, Colombia, 2015 to 2024. Panel A, crude ratio; panel B, global empirical Bayes smoothed ratio, both on the same scale. Crude rates are the input for the spatial autocorrelation tests. On the shared scale panel B appears uniform because the smoothing pulled every municipal rate into a narrow band from 65.7 to 67.3 per 100 000, consistent with between-municipality variation compatible with chance.



Supplementary Figure S4. Funnel plot of the municipal maternal mortality ratio against live births, Córdoba, Colombia, 2015 to 2024. The horizontal line is the departmental ratio and the curves are the 95% and 99.8% limits computed with the normal approximation to the Poisson distribution.



Supplementary Figure S5. Municipality of occurrence of maternal deaths and deaths crossing departmental boundaries, Córdoba, Colombia, 2015 to 2024. Panel A, where the 162 deaths of women resident in the department occurred. Panel B, deaths of women resident elsewhere that occurred in the department and deaths of its residents that occurred elsewhere.